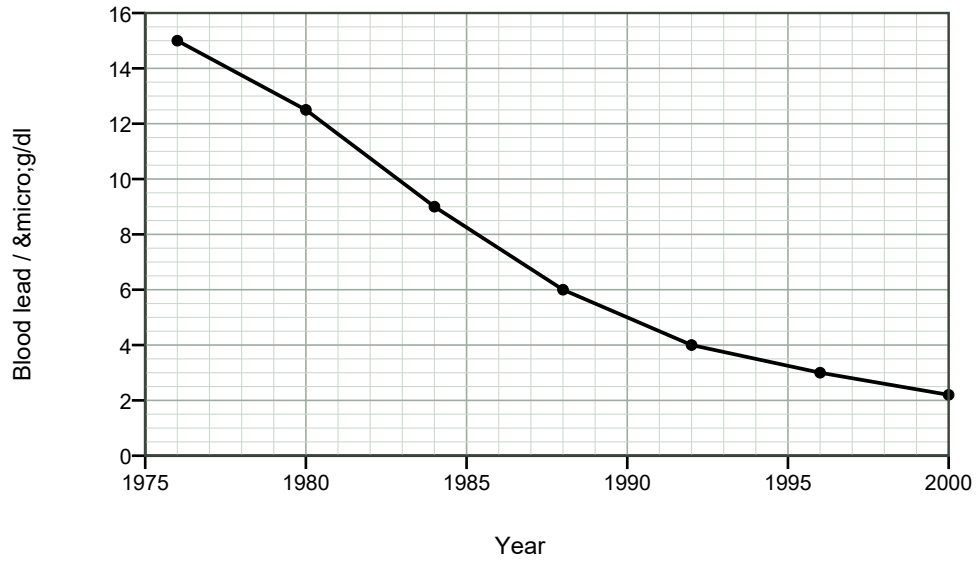
Questions by topic: Heavy metals: lead, mercury - Set 1 (55 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 1 . 1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer ____ %

0 1 . 2 Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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0 1 . 3 Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

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0 1 . 4 Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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0 1 . 5 Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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0 2 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 2 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

0 2 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

0 2 . 3 Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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0 3 . 1 Give **three** sources of mercury pollution.

[3 marks]

1 .....

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2 .....

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0 3 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

1 .....

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2 .....

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3 .....

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0 3 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 3 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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0 4 Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg^{-1}	Cadmium in kidney of residents / mg kg^{-1}	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

0 4 . 1 Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer ..... times

0 4 . 2 Explain why cadmium and zinc pollution often occur together.

[1 mark]

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0 4 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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0 4 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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0 4 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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0 5 . 1 Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour ☐

B Children pick up dust / flakes and suck their fingers ☐

C Lead is more soluble in children's blood ☐

D Children spend more time near roads ☐

0 5 . 2 Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

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0 5 . 3 Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

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0 5 . 4 Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

1 .....

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0 5 . 5 Explain why heavy metals are described as neurotoxins.

[1 mark]

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0 6 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 6 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 6 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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END OF QUESTIONS

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